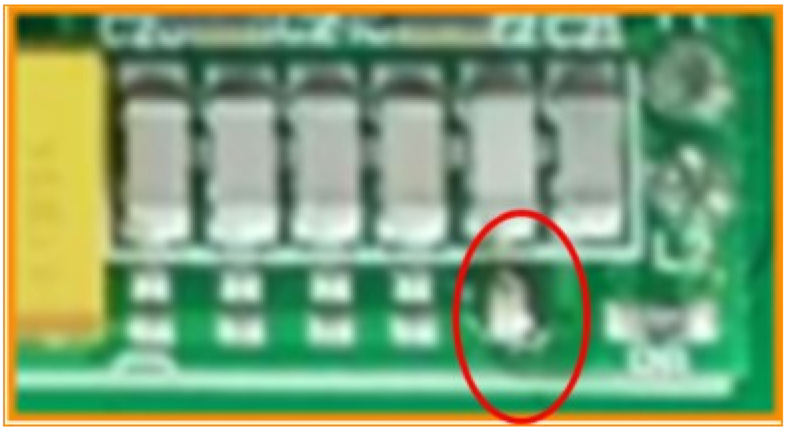
Antenna Topics:

Q: For the capacitors used to 'tune' antennas, the paper refers to them as' J1 to J5 Capacitors'. The image on the page shows the ratings of six (6) capacitors instead of five. So, are there really five or six?
A: There are 6 capacitors on the module, but the right most one, near the antenna connection points ANT1/ANT2, is fixed and cannot be changed. The five on the left can be added to the circuit or removed (erased) from the overall capacitance of the card.

Q:"Delete all jumpers" is written on the page. How do we erase all jumpers?
A: Using an electric soldering iron to erase requires a certain level of skill in using an electric soldering iron. There are soldering pads directly underneath them that can have a jumper soldered across them to add capacitance to the circuit as shown in the following, fuzzy, picture.



Q: Why do I need to change the resonant capacitance of the antenna?
A: For Load Coil (LC) resonant circuits, slight changes in inductance or capacitance parameters are sufficient to have a significant impact on the resonant voltage on the antenna. When the antenna is installed in place, it is already in a different state than when it is suspended, especially when there are other metals or circuit boards near the antenna. Another reason is that the fixed main resonant capacitor on the module is 2.2nF, which is to the right of the J5 capacitor and cannot be removed from the circuit, and the available high voltage withstand capacitance is NPO 5% accuracy. This 5% production capacitance error is also sufficient to affect the card reading distance. Therefore, five small capacitors are needed to compensate for the capacitance of the main resonant capacitor to fine tune the resonance to achieve the best card reading effect. The capacitance of the WL-134, as delivered, is tuned with the provided antenna. If you make changes to the antenna or use your own design, you will need to perform the 'capacitance tuning' process to have it work with the interface card in order to read RFID chips.

Q: The instruction sheet says, "The antenna has insulating paint. It must be welded to the end of the copper wire". How is this "welding" completed and how do you handle the insulating paint on the wire?

A: The two antenna wires have had the insulating paint removed from them so they can be soldered to the ANT1 and ANT2 thru holes using an electric soldering iron. Be careful when soldering so you do not overheat nearby components and damage them or the circuit board.

Q: The ANT1 and ANT2 connections on the interface card are very close to each other. If the two wires of the 97mm x 97mm square antenna contact each other during the connection to the interface card, is there a problem?
A: Be careful not to short-circuit by having the unpainted ends from the antenna touch one another, which may damage the resistance on the module.

Q: Has the capacitor on the interface card been "tuned" to the supplied square antenna? I hope I can achieve the optimal reading distance between the antenna and the RFID chip without making any changes to the card or the antenna. At this point, I will use a 9-volt battery to power it.
A: Yes, it has been adjusted to an ideal state before leaving the factory. You can directly connect to a 9V alkaline battery. If a 1000uF capacitor can be connected in parallel at the battery output terminal, the card reading effect will be improved.

Q: The manual says "ANT2 340V VPP". Is there really such a high voltage at that connection point?

A: Yes, the resonance voltage on the antenna is so high that it will be higher in good resonance. There is a sine wave of 134.2K on the antenna. When the module is working, do not touch the pad at the antenna ANT2 with your hand. Although the current is small, it will feel bad to touch.

Q: Can I make my own antenna that suits my application?
A: Yes, use any wire or 0.3mm enameled wire to wind a hollow coil suitable for your size, with an inductance of 580uH. Then fine-tune the resonant capacitance combination of the module to maximize the card reading distance. You will need an inductance meter. If you don't have an inductance meter available, it can be a bit troublesome. You can wrap the coil around the module during operation, starting with approximately 42 turns (10X10CM SIZE), and then observe the module's operating current, trying to increase or decrease the coil until the module's operating current is near the recommended value. Reference operating current 80mA@6V, 110mA@9V.

Q: Why is the provided antenna so tightly wound, by what method and how is it held together?
A: The provided antenna is automatically produced and wound by a wire winding machine, with a dedicated antenna mold. The enameled wire used is either hot air or acetone self-adhesive enameled wire. During production, the heated air will make coil wire stick together, so that the wires of antenna will remain in the shape you see.

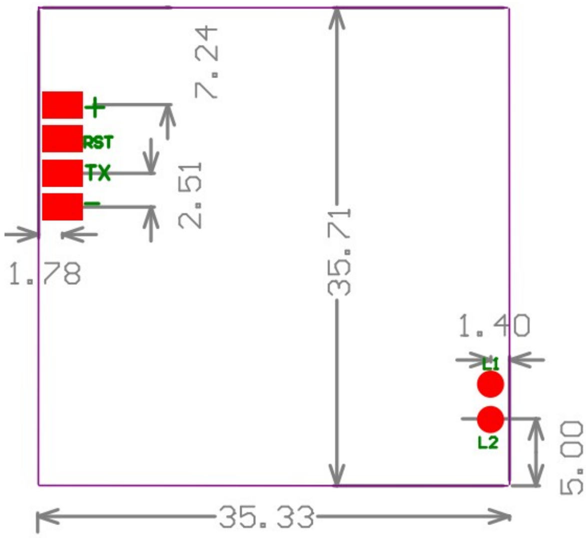
Q: The attached antenna is a bit too large for me and the shape is not suitable. Can I make it smaller?

A: You cannot arbitrarily change the shape of the antenna. Changing the shape of the antenna can change the inductance of the antenna, which can seriously affect the card reading effect. If you need other antennas of different sizes or shapes, please contact us.

General Interface Card Questions:

Q. Can the module interface card be made larger so that there is at least one mounting hole that can be used to physically mount the card in place?

A: In most cases, modules are welded to the customer's motherboard using the thru holes for VCC/TX/RST/Ground and ANT1/ANT2. The following diagram has the dimensions of the interface card and shows the offset locations of those six (6) thru holes so that you can make your own 'motherboard' to mount the WL-134 interface card to and design that motherboard card to have the mounting holes you want, where you need them.



Programming And Interface Topics:

Q: Do you have any library files or code suggestions for Arduino usage?

A: Yes. Please refer to Arduino library entry of: Rfid134 LIB_by_Makuna-1.0.2. If you examine the RFID134.h file, you will notice a method of converting hexadecimal ASCII characters into binary numbers and then combine them into an 'unsigned long long' (64 bit) variable. There are several examples of how to print out 'long long' Arduino variables at: https://stackoverflow.com/questions/33947988/arduino-how-can-i-print-the-unsigned-long-long-data

Q: What is meant by 'LSB First' on the instruction sheet for the data we received from the TX port of the interface module?

A: The 'card number' within the data stream, read from the TX port, is bytes 1 through 10 while the country code is bytes 11 through 14. What 'LSB First' means is that when you start processing the data for either value, you **start** with byte 10 and work down to byte 1 for the card number and start at byte 14 and work down to byte 11 for the country code. Remember that byte 0 is the leading '02' value indicating the start of a RFID chip data stream.

Q: What is the RESET (RST) pin used for?
A: RST is used to reset the interface module. In some cases, the customer wishes to continuously output the card number. At this time, an IO port, such as a PWM pin on an Arduino microcontroller, can be used to control the interface module RST pin. Each time RST is lowered for 10ms and then raised, the interface module will reset and reread the card output card number. For example, if the interface module is reset four times per second, the module will output the RFID chip ASCII string four times. Repeated resetting of the interface module will not have any other side effects. If not in use, please suspend, leave the pin disconnected. **Note that the RST pin is at a 5V TTL level, do not directly connect to the IO port of a 3.3V microcontroller**

Q: What is the purpose of the six (6) bytes of 'User data' within the 30 byte data stream as mentioned on the instruction sheet?
A: These User Bytes can contain any data specific to an application or usage. If you have access to a RFID card writer capable of writing you own data to these six bytes, it is possible to write HEX data to the data area of the RFID data stream and read it out through the interface module. There is a kind of temperature tag that expresses temperature through the user bit, for example.

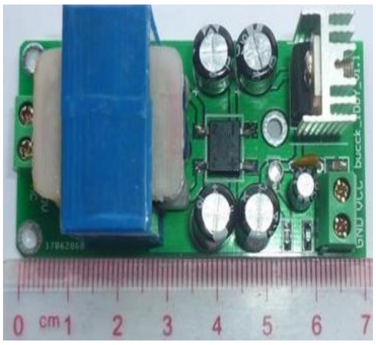
Q: I want to make a pet automatic feeder or pet door. How should I design it?
A: You can use a single chip computer to connect an infrared radiation sensor. When an animal blocks the infrared sensor, start the RFID module power supply, read the RFID tag on the animal, and compare it with the card number stored in the EEPROM of the single chip computer. If the card number is correct, use the single chip computer to control the motor action, feed the animal, or open the door. Remember to use a linear power supply or a switching power supply connected to AC ground. If you use battery power, the main problem is low power consumption. You can hibernate the microcontroller and turn on the infrared sensor for 5 ms every second to check for animals.

Power Supply Topics:

To start this topic off, it should be understood that in order to 'energize' the RFID Microchip, that has been embedded within an animal, and have the data encoded within it be able to be read **requires highly tuned** radio waves on the 134.2kHz band! If there is other devices that are close by that also generate radio waves that are in **close** proximity to 134.2kHz they will interfere with the radio waves that the WL-134 device generates and uses to read the RFID data. One general device that can generate radio waves that can interfere with the WL-134 is a switching power supply.
The definition of 'close proximity' to the WL-134 device is within **1** feet.

With that in mind, here are power supply Q&A's.

Q: Why does switching power supply seriously interfere with card reading?
A: The common switching power supply operating frequency is between 50KHZ and 150KHZ, which coincides with the operating frequency of the 134.2K card reader module. There is a common mode electrical wave 'noise' between the positive and negative electrodes of the switching power supply, which cannot be filtered by simple large capacitance or inductance. When the power supply is connected to AC ground, it can have a positive effect on filtering out this common mode noise. Compared to switching power supplies, linear power supplies are much better in terms of reduced radio frequency (RF) interference. Another option is to use switching power supplies with switching frequencies above 200KHZ, but such switching power supplies are not commonly used and are also expensive.

linear power supply (fixed input voltage 110V or 220V,There is a large transformer)



Switched mode power supply (input voltage 100V~240V)

Q: I am using a 12V switching power supply or a mobile phone charger. Is it effective to add a linear voltage regulator such as 7809 and a large capacitance filter between either of these power supplies and the WL-134?
A: The input is a switching power supply, and the power input already has common mode noise, which cannot be filtered out by linear regulators such as 7809 and large capacitors, unless the module GND is connected to AC ground.

Q: "The AC power sockets here are all two-hole sockets. How should I connect to the AC ground?"

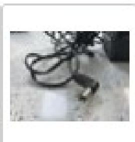
A: In some places, AC power outlets do not have a ground hole, so it is necessary to use a linear power supply or high frequency switching power supply or connect a large ground wire by yourself. This is true grounding, and there is a dedicated grounding pin. The pin should be buried underground, preferably in a humid environment. If you cannot ground and cannot buy a high frequency switching power supply, you need to design your own common mode filter, which can be quite troublesome. The following are pictures of some 'grounding bars' that would be driven into the group and attached to the GND pins of the WL-134 and any computer attached to it.

Q: Where can I buy a linear power supply?
A: Linear power supplies are usually heavier and have larger enclosures, because there is a large transformer inside. If you do web searches using the search terms of "9 volt DC power supply linear" you will get a lot of matches that you can go through to find a power supply that meets the requirements. Specifically look for the term 'linear' in the product description. There should be no mention, reference or inference to 'switching'. You can always send and email or call the vendor to specifically ask if the power supply is linear.

One such supply is this one:

https://www.amazon.com/-/zh/dp/B00B886CWS/ref=sr_1_90?crid=1NTPB1OWMXP4J&keywords=linear+power+supply+9v&qid=1680093645&sprefix=%2Caps%2C303&sr=8-90






Q:Can AA batteries provide stable electricity well? 4x1.5V or 6x1.5V?

A:AA batteries are ideal, 4X1.5V would be great, but using 6X1.5V would be a bit troublesome. In order to save battery costs, some customers have used 2X3.7V (7.4V) lithium batteries for power supply, with very good results.

Q:If powered by USB C, will there be a difference between increasing 5V or activating 9V mode on USB PD? Or does it depend on the battery pack/power supply?
A:If USB C is used for power supply, there is not much difference between 5V and 9V. The key lies in how 5V is obtained. I have sent you detailed information about the module, which includes special instructions on the power supply.

Q:If you were to create a handheld scanner and need to extract 5V voltage for Pico and power the module as much as possible, what would you do?
A: I will use two 3.7V lithium batteries in series, and then use LM7805 to reduce the voltage to provide 5V to PICO and 7.4V to power the card reading module. This solution is very simple and the card reading effect is very good. A charger for a 7.4V 2S lithium battery can be purchased

Q:Does the WiFi or Bluetooth radio on Pico interfere with reading distance?
A:Yes, but it's also easy to solve. By using a 10R+470uF RC filtering circuit, most of the interference can be filtered out.


HUAYPG 专注高品质电池
18650锂离子电池 Lithium-ion Battery
□2000 □2200 □2500
□3000 □3200 □3400
mAh
7.4V
标称电压:7.4V 充满电压:8.4V
警告&免责申明：请一定要比对原装电池弄清楚(红+)(黑-)线端子排列顺序。
因极性接入错误或用错电压引起机子/设备损坏，后果买家自负哦。

7.4v BANLANCE CHARGER